

STATEMENT OF REQUIREMENTS

TITLE: QUANTUM SENSORS MILLIMETER-WAVE VECTOR NETWORK ANALYZER

I. BACKGROUND INFORMATION

The Quantum Sensors Division (687) of the Physical Measurement Laboratory (PML) of NIST conducts state-of-the-art development and characterization of millimeter-wave quantum sensors and optics to support the most sensitive millimeter-wave research done in the world. Direct, coherent millimeter-wave measurements require a low-noise vector network analyzer (VNA) capable of operation up to 20 GHz with the appropriate ports, sources, and measurement capabilities to operate and measure network S-parameters at frequencies up to 1.1 THz using existing frequency extenders (often called VNA Extenders). NIST's associated VNA is nearing its end of life and requires replacement. NIST does not require new frequency extenders, so a drop-in replacement for the VNA is required to avoid significant additional costs and delays.

II. PURPOSE

NIST requires a drop-in replacement vector network analyzer (VNA) that can operate and perform measurements from existing high frequency equipment. To accomplish mission goals utilizing high sensitivity NIST sensors, this VNA must meet the outlined specifications.

PLACE OF PERFORMANCE

NIST Boulder: 325 Broadway, Boulder, Colorado, Building 1, room 1-2120.

Regular Business Hours

N/A – Required device and parts can be shipped.

PERIOD OF PERFORMANCE

Delivery 24 weeks ARO.

III. SPECIFIC REQUIREMENTS

The VNA Must:

- Be a Keysight VNA Model N5222B-419 with additional options “020” (IF Inputs) and “UNY” (Enhanced low phase noise), or equivalent
- Be drop-in compatible with NIST-Boulder existing hardware including but not limited to VNA extenders up to 1.1 THz.
- Be drop-in compatible with NIST-Boulder existing software package written for interfacing Keysight VNAs
- Be provided by Keysight (the original equipment manufacturer, OEM) or an authorized OEM reseller while maintaining OEM calibration and warranty.
- Either factory new or a unit refurbished to OEM like-new specifications or better, is technically acceptable. If the unit has been refurbished, the refurbishment shall use OEM parts with refurbishment performed by OEM factory trained personnel, and have a new and valid OEM calibration after refurbishment. The Government has a preference for units refurbished to OEM like-new specifications or better.

Performance requirements of the N5222B-419, or equivalent. The unit shall:

- Perform all standard functions of a VNA as per common in the scientific and industrial capacity including frequency sweeps of network S-parameter measurements with the ability to store and transfer measurement data by common electronic means and display the data on the unit's integrated screen.

- Have a phase noise specification on two or more ports of:
 - -115 dBc/Hz or better using an internal source at 20 GHz with 10 kHz offset
 - -130 dBc/Hz or better using an internal source at 1 GHz with 10 kHz offset
- Operate at base frequencies up to 20 GHz or higher.
- Operate at base frequencies as low as 100 MHz or lower.
- Have 4 VNA ports that operate at the full frequency range.
- Have at least two internal frequency sources that are phase referenced to each other
- Maximum source powers +13 dBm, or higher, on at least two ports
- Have front panel Intermediate Frequency (IF) port access.
- Have front panel digital display screen and control buttons
- Have the ability to be controlled remotely from a computer via software (an Application Programming Interface (API)) using common industry protocols.
- Dynamic range of 125 dB, or larger, at 18 GHz on all ports.
- Include a 1-year OEM warranty on unit and parts.
- Be electrically powered by US standard 120V / 60Hz

The unit must be in compliance with the following Government information technology standards and the contractor must:

- Ensure the unit is compliant with DOC [Enterprise Cybersecurity Policy \(ECP\) Program](#) requirements
- Demonstrate conformity using the USGv6 Suppliers Declaration of Conformity ([USGv6 Profile Supplier's Declaration of Conformity | NIST](#))

IV. GENERAL REQUIREMENTS

Inspection & Acceptance

In addition to the inspection and acceptance terms articulated in the specific FAR clause that allows the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

Performance Tests:

1. NIST will confirm unit compatibility and performance is to both the requirements listed above and those published by the manufacturer for that VNA model within 30 days of delivery.

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST POC to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within 30 days of the receipt of the equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the

Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

REQUIREMENTS FOR CONTRACTOR PERSONNEL

If the Contractor wishes to have an employee present during acceptance testing, the contractor accepts that said employee must adhere to the following:

Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

V. SCHEDULE OF DELIVERABLES

Define all deliverables and indicate (1) the quantity desired; (2) the format in which each deliverable is to be supplied; (3) the delivery date (based on the number of weeks/days); (4) the place of delivery. The context of each deliverable shall be clear. **Keep in mind that every task does not require a deliverable. Deliverables are always tangible items (i.e., report, minutes, etc.)**

Complete the table providing the information for all the tasks identified above.

<i>Deliverable Number</i>	<i>Description</i>	<i>Format</i>	<i>Due Date</i>	<i>Quantity</i>
1	Vector Network Analyzer	Physical stand alone unit	24 weeks after award	1

Standards of Acceptance: The NIST POC or COR shall review all the above deliverables and respond with an acceptance or request for revision email to the Contractor Point of Contact (POC) within 30 days of receipt of deliverable.

VI: NIST Contacts

- Primary Technical Point of Contact: Jason Austermann
- NIST COR: Michelle Chaumont